System Design (Telusko) - Day 13

One-Shot 5h course | Video 3:00-3:15

Consistent Hashing

Distribute keys across nodes so adding/removing nodes moves little data.

Key concepts

- Naive hash(key) % N remaps almost everything when N changes.
- Consistent hashing maps nodes and keys onto a ring.
- Virtual nodes balance the load evenly.
- Used by caches, sharded DBs, and CDNs.

Key example

On the hash ring, a key goes to the next node clockwise; adding a node moves only its arc.

Today's tasks (1.5h)

[ ] Learn the hash ring
[ ] Explain why % N is bad
[ ] Note virtual nodes

Resources

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Video:  youtu.be/Vnm-ycSfJx4   (start at 3:00-3:15)
Docs:   docs.telusko.com/docs/system-design
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(Full clickable links are in the app: Goals > this day > Study block.)